

Multiple Choice (10 marks)

1. A chain of mass m per unit length and total length l lies in a pile on the floor. At time $t = 0$ a force P is applied and the chain is raised with constant velocity v . Find $P(t)$.
 - (a) $P = mv(t + g)$
 - (b) $P = m(v + gt)^2$
 - (c) $P = mgvt + mv^2 = mv(v + gt)$
 - (d) $P = mv^2 - mgvt$
 - (e) $P = mv + gt$
2. How much heat is produced in 5 minutes by an electric iron which draws 5 amperes from a 120-volt line?
 - (a) 72,000 calories
 - (b) 80,000 calories
 - (c) 36,000 calories
 - (d) 65,000 calories
 - (e) 48,000 calories
3. The two moons of Mars are called ...
 - (a) Deimos and Triton
 - (b) Phobos and Demos
 - (c) Triton and Deimos
 - (d) Phobos and Deimos
 - (e) Tritos and Desmos
4. The electric flux through a spherical surface is $4.0 \times 10^4 N \cdot m^2/C$. What is the net charge enclosed by the surface?
 - (a) 2.65 e-07
 - (b) 7.10 e-07
 - (c) 2.30 e-07
 - (d) 8.45 e-07
 - (e) 5.67 e-07
5. What is the illumination 5 feet from a 75 candlepower lamp?
 - (a) 5 foot-candles

- (b) 1 foot-candle
- (c) 50 foot-candles
- (d) 15 foot-candles
- (e) 20 foot-candles

True / False (10 marks)

Answer True or False

6. True or False: The force applied by the pitcher to the baseball is 10 pounds, assuming a direct proportionality between weight and force without considering acceleration.
7. True or False: The magnifications for an object at 20 mm, -100 mm, and +100 mm with a 50 mm lens are $(4/3)$, $(1/5)$, and 0, respectively.
8. True or False: The work done in dragging a 150 kg mass over 10 meters with a coefficient of sliding friction of 0.30 is 5000 joules.
9. True or False: The focal length of the lens in a slide projector designed for 35-mm slides projecting a 40 in by 60 in image 10 ft away is 2.69 inches.
10. True or False: Natural boron has an atomic weight of approximately 11, reflecting a simple average of its isotopes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Consider Sally driving a 1965 Mustang along a straight highway with an x-acceleration described by the equation $a_x = 2.0 \text{ m/s}^2 - (0.10 \text{ m/s}^3) t$. Analyze the conditions under which the car reaches its maximum x-velocity and calculate the corresponding distance X traveled when this occurs.
12. Calculate and discuss the total heat energy required to convert 25 kg of ice at -10°C to steam at 100°C, detailing each phase change and the associated calculations.

End of Paper